INCH-POUND

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MILITARY SPECIFICATION

ICE MILK, MILK SHAKE AND YOGURT MIXES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers soft serve ice milks, soft serve yogurt mixes, direct draw milk shake mixes, and dual purpose dehydrated mix which can be reconstituted with water and frozen to provide either soft serve ice milk, or direct draw milk shakes for use by the Department of Defense as items of general issue (see 6.1).

1.2 Classification. The product shall be of the following types, classes, and styles as specified (see 6.1).

Type I - Fresh soft serve ice milk mix
  Style A - Vanilla flavored
  Style B - Chocolate flavored

Type II - Fresh direct draw milk shake mix
  Style A - Vanilla flavored
  Style B - Chocolate flavored

Type III - Dual purpose dehydrated soft serve mix
  (for soft serve and direct draw milk shakes)
  Class 1 - Regular mix containing milk fat
  Class 2 - Imitation mix containing vegetable fat
  Style A - Vanilla flavored
  Style B - Chocolate flavored

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8910

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Type IV - Fresh soft serve yogurt mix (low fat)
Class 1 - Regular mix containing milk fat
Class 2 - Imitation mix containing vegetable fat
Style A - Vanilla flavored
Style C - Fruit flavored
Black cherry, Black rasberry, Blueberry, Cherry, Lemon, Orange, Peach, Red rasberry, or Strawberry, as specified (see 6.2)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-C-495	- Coatings, Exterior, for Tinned Food Cans
PPP-C-29	- Canned Subsistence Items, Packaging Of
PPP-C-96	- Cans, Metal, 28 Gage and Lighter
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-1163	- Box, Corrugated Fiberboard, High Compression Strength, Weather-Resistant, Wax-Resin Impregnated

MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-F-35081	- Fat, Hydrogenated, Vegetable

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-668	- Sanitary Standards for Food Plants

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR, Parts 1-199, 7 CFR, Parts 58.135 through 58.138, 9 CFR 77, and 9 CFR 78).

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

Bacteriological Analytical Manual

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. DEPARTMENT OF AGRICULTURE

U.S. Standards for Grades of Butter

Dairy Plants Surveyed and Approved for USDA Grading Service

Methods of Laboratory Analysis

General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products

Milk for Manufacturing Purposes and Its Production and Processing

U.S. Standards for Grades of Dry Buttermilk

U.S. Standards for Grades of Dry Whole Milk

U.S. Standards for Dry Whey

(Copies are available from the Dairy Standardization Section, Dairy Division, Agricultural Marketing Service, Room 2750, South Building, P.O. Box 96456, US Department of Agriculture, Washington, DC 20090-6456.)

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ENVIRONMENTAL PROTECTION AGENCY

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INTERNATIONAL ASSOCIATION OF MILK, FOOD, AND ENVIRONMENTAL SANITARIANS, INC.

Applicable 3-A Sanitary Standards and 3-A Accepted Practices

(Application for copies should be addressed to the International Association of Milk, Food, and Environmental Sanitarians, Inc., P.O. Box 701, Ames, IA 50010, Telephone 1-800-525-5223.)

AMERICAN PUBLIC HEALTH ASSOCIATION

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1015 15th Street, N.W., Washington, DC 20005.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN DAIRY PRODUCTS INSTITUTE

Standards for Grades of Dry Milks, Including Methods of Analysis, Bulletin 916

(Application for copies should be addressed to the American Dairy Products Institute, 130 North Franklin St., Chicago, IL 60606.)

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AMERICAN OIL CHEMISTS' SOCIETY

Official Methods and Recommended Practices of the American Oil Chemists Society

(Application for copies should be addressed to the American Oil Chemists' Society, P.O. Box 5037, Station A, Champaign, IL 61820.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1120, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.1.1 Raw milk. The commingled raw milk used in the preparation of this product shall comply with all of the requirements for acceptable milk as specified in the General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products, (7 CFR Parts 58.135 through 58.138). When Grade A milk products are required to be used for type IV product by state or local regulations, such requirements shall be met. Undergrade milk shall not be used. The milk shall be drawn from cows located in a Modified Accredited Area or Accepted Free Herd for Tuberculosis as outlined in 9 CFR 77. In addition, the milk shall be from cows located in states meeting class B or better, or in an area involved with milk ring testing or a blood testing program for brucellosis as outlined in 9 CFR 78. The milk shall be practically free from colostrum; shall be fresh, wholesome, and normal in appearance and odor; and shall be subject to inspection by the procuring agency or its duly authorized representative. The commingled raw milk, unless processed within 2 hours after

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receipt shall be cooled immediately to 45°F or below and maintained at that temperature until separation and /or processing is started. The raw milk shall be used within 60 hours following receipt.

3.1.2 Dairy and dairy derived ingredients. The dairy and dairy derived ingredients shall comply and be used in accordance with all regulations cited in 21 CFR Part 135 for Frozen Desserts and, as applicable 21 CFR Part 131 for Milk and Cream as promulgated by the Federal Food, Drug, and Cosmetic Act. In addition they shall meet the requirements specified in 3.1.2.1 through 3.1.2.5.

3.1.2.1 Skim milk, partially skimmed milk, sweet cream buttermilk, and cream. Skim milk, partially skimmed milk and cream shall be obtained by the separation of whole milk conforming to the raw milk requirements of 3.1.1. The raw skim milk, partially skimmed milk, and cream shall be maintained at a temperature not higher than 45°F and shall be used within 60 hours following the receipt of the raw milk from which it was separated. Pasteurized skim milk, partially skimmed milk, sweet cream buttermilk, and cream shall not exceed a standard plate count of 50,000 per mL or gram (as applicable), shall be maintained at a temperature not higher than 45°F, and shall be used within 96 hours following receipt. Pasteurized cream may be frozen and maintained at a temperature not to exceed 0°F and shall be no older than 120 days at time of use.

3.1.2.2 Concentrated fluid dairy products and other dairy derived ingredients. Concentrated fluid dairy products and other dairy derived ingredients to increase the nonfat milk solids content shall possess a sweet pleasing milk flavor, free from objectionable off-flavors and odors, such as fishy, rancid, oxidized or scorched. The standard plate count shall not exceed 50,000 per gram. The product shall be maintained at a temperature not higher than 45°F and shall be used within 96 hours following receipt.

3.1.2.3 Milkfats and oils. Anhydrous milkfat, butter oil, plastic cream, and churned fat shall meet the finished product and analytical requirements of the General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products.

3.1.2.4 Butter. Butter shall conform to the requirements for U.S. Grade A or better at time of use and as defined in the U.S. Standards for Grades of Butter. The butter shall be unsalted.

3.1.2.5 Dry milk products. Dry milk products, such as nonfat dry milk, dry buttermilk, dry whole milk, and dry whey, shall be manufactured by the spray process and shall conform to the requirements for U.S. Extra Grade at the time of use and as defined in U.S. Standards for Grades for the product specified. The product shall have been manufactured in a plant listed in Dairy Plants Surveyed and Approved for USDA Grading Service. When Grade A milk products are required to be used for type IV product by state or local regulation, such requirements shall be met.

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3.1.3 Vegetable fats. Vegetable fat when used for class 2 product (see 6.2) shall conform to the requirements of MIL-F-35081 except that the Wiley melting point shall be 77° ± 2°F. Alternatively, coconut fat conforming to the following requirements may be used:

a. The flavor shall be bland.
b. The product shall be free from extraneous matter and the liquified product shall be clear and brilliant.
c. The color shall be 15 yellow and 1.5 red maximum (Lovibond).
d. Moisture and volatile matter shall not exceed 0.1 percent.
e. The melting point determined by the Wiley method shall be 93° ± 3°F.
f. The peroxide value shall not exceed 1.0 meq./kg of fat.
g. The free fatty acids shall not exceed 0.05 percent (as oleic).
h. The iodine value shall not exceed 9.0.
i. The fat stability (AOM) shall be not less than 100 hours.

3.1.4 Sweeteners. Nutritive carbohydrate sweeteners shall comply with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.1.5 Vanilla flavoring. Vanilla flavoring shall be ground vanilla beans (vanilla powder), vanilla extract, imitation vanilla extract, synthetic ethyl or methyl vanillin, or combinations thereof.

3.1.6 Chocolate and cocoa products. Chocolate and cocoa products used as flavoring shall meet the requirements permitted in the Definitions and Standards for Cocoa Products.

3.1.7 Fruits and fruit flavorings. Fruits, preserves, fruit puree, natural fruit extracts, concentrates and other flavoring material shall be in their natural state, heat processed, dehydrated or frozen, and especially adapted for use in frozen yogurt. Fruit flavors that are used shall impart a clean, natural, and pleasing flavor reaction and shall be characteristic of the applicable fruit. Imitation flavors shall not be used.

3.1.8 Stabilizers and emulsifiers. Colloidal stabilizers, stabilizing salts, and emulsifying agents shall comply with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.1.9 Vitamin A. Vitamin A palmitate, used in type III, styles A and B products, shall comply with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.1.10 Water. Water used for formulation shall conform to the National Primary Drinking Water Regulations.

3.1.11 Packaging gas. Packaging gas shall be of food grade quality and shall consist of pure nitrogen or a mixture of nitrogen and approximately 10 percent carbon dioxide plus other inert gases of the atmosphere, but shall contain no more than 0.5 percent oxygen.

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3.1.12 Yogurt culture. Yogurt culture (used in type IV product) shall be reliable, high activity, 1:1 blend of Lactobacillus bulgaricus and Streptococcus thermophilus organisms and shall be capable of producing a product meeting the finished product requirements of 3.3 (see 6.1).

3.1.13 Coloring material. Coloring material shall be natural or artificial, providing it conforms to the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. Color shall be used only when compatible with the specific flavor and to enhance the natural color of the flavoring used if lacking in shade, hue, or intensity.

3.2 Processing. Processing shall be in accordance with the following requirements and 3.2.1:

a. All raw milk, skim milk, partially skimmed milk and cream shall be at the temperature and within the age requirements specified in 3.1.1 and 3.1.2.1 at time of use and shall be handled and processed in accordance with Good Manufacturing Practices.

b. All types of mixes shall be formulated by using the applicable ingredients specified in 3.1 in the quantities normally used in recognized commercial practice provided the finished product meets the applicable requirements of 3.3 and 21 CFR 135 and the exceptions specified in 3.1.2.1 and 3.2.1 (also see 6.1 for recommended guidance) are met. Milk fat and vegetable fat (other than that supplied by the chocolate flavor or cocoa) shall not be used in the same mix. For type III product, all or part of the dry sweetener, vanilla flavoring material, cocoa, cold water soluble stabilizers, and vitamin A may be dry blended with the spray dried product after it is processed according to 3.2c-g. Optionally, the stabilizers or emulsifiers may be dry blended provided that this material is water-soluble, the finished reconstituted product complies with the requirements of 3.3, and in no way affects the finished product for its intended purpose.

c. Whey products, corn syrup solids, and emulsifying agents shall not be used in type II product formulations. Buttermilk solids shall not be used in type III products.

d. The entire mix, after proper formulation (to include chocolate or cocoa flavoring, when applicable, except those ingredients specifically allowed to be dry blended in type III product), shall be pasteurized by one of the following methods:

(1) by heating every particle to at least 155°F and holding at this temperature for at least 30 minutes;
(2) by heating every particle to at least 175°F and holding at this temperature for at least 25 seconds; or
(3) any other method of pasteurization recognized as equally efficient by the Milk Safety Branch, Food and Drug Administration, U.S. Department of Health and Human Services and approved by the appropriate state or local authority.

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The pasteurized mix shall yield less than 1 microgram of phenol per gram when tested in accordance with the Scharer Rapid Phosphatase Method. (Note: The phosphatase test shall be considered a screening procedure and in no way shall a negative test be considered as positive proof of proper pasteurization as defined by the appropriate health authority. A positive test shall be cause for immediate follow-up action to determine if improper pasteurization is directly involved.) All multi-use containers, equipment, and appurtenances with which dairy or other food products come into contact shall be of approved sanitary design and construction. Equipment for which there are 3A Sanitary Standards and Accepted Practices shall conform to such standards as accepted practices.

e. The mix shall be homogenized effectively in accordance with good industry practice to prevent fat separation. Powdered fat or shortening products are not permitted in type III, class 2 product.

f. The mix shall be cooled immediately following the pasteurization/ homogenization processing to a temperature of 45°F or below and held at this temperature while being filled into an appropriate package, and during storage if the product is to be delivered within 24 hours (see 3.2g for type III product and 3.4 for exception).

g. During any of the processing steps prior to spray drying for type III product, if there is any delay longer than 30 minutes in the continuous flow of the fluid mix, it shall be cooled to a temperature of 45°F or below and maintained at that temperature until processing resumes. If processing delays exceed 24 hours, the product shall be unacceptable.

The dehydrated product shall be produced by spray drying, except where dry blending is allowed, to not less than 98 percent total solids. Spray dryers shall be of the continuous product removal type. Wet collectors may be used provided they are of sanitary design approved by applicable regulatory authorities, that they do not recycle the mix, and that they are operated under such conditions so as to ensure that the end product meets the requirements of 3.3. The appropriate vitamin A palmitate shall be blended with the dehydrated product at a level of 11,200 I.U. per pound of dry mix.

3.2.1 Processing of type IV product. The basic dairy products and nondairy ingredients shall be formulated, standardized, and blended in accordance with good commercial practice. (See 6.1 for guidelines with respect to recommended procedure.) The blend shall be subjected to high temperature heat treatment (180°F to 190°F) for 30 minutes or its thermal equivalent, homogenized as in 3.2e. Following homogenization, the product shall be cooled promptly to incubation temperature (105°F to 110°F) or any combination of time and temperature yielding equivalent results.

3.2.1.1 Ripening and further processing. The product shall be inoculated with the yogurt culture in sufficient quantity to permit the proper development of lactic acid and the characteristic yogurt flavor (acetaldehyde) and to meet the end item requirements of 3.3 (see also 6.1). A preliminary cooling period

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to below 80°F should follow before final additions of sweetener and flavors/ coloring are made in order to prevent wheying-off. Complete cooling to 40°F should follow promptly to control and prevent the development of excess acidity. In the event the product is coarse, grainy, or lumpy, a texturizing procedure shall follow to render the product smooth. The addition of fruit or preserves shall follow texturizing. If the finished product is to be shipped in frozen form, freezing to -15° to -20°F shall be accomplished within 48 hours.

3.3 Finished product requirements. Types I, II, and III (after reconstitution to a soft serve fluid mix (see 5.4.1), shall have a clean, pleasant, creamy, moderately sweet flavor, characteristic of the vanilla or chocolate flavor types used in commercial operations; and shall be classified and labeled in accordance with Definitions and Standards of Identity for Frozen Desserts (see 3.7). A mildly cooked flavor shall be considered acceptable. After reconstitution, the type III product shall be free from objectionable flavors and odors such as oxidized, stale, rancid, tallowy, scorched, or fishy. The body of the liquid mixes, types I, II, and III (reconstituted), shall be moderately viscose and smooth, equivalent to that of fresh, heavy sweet cream. The viscosity shall be such as to be readily flowable from the mix hopper of the soft serve or milk shake type freezer into the freezing cylinder. Such defects as ropy, curdled, wheyed-off, watery, foamy, doughy, etc shall be unacceptable. The liquid mix shall be free from "fish eyes" or the evidence of undissolved stabilizer or emulsifier suspended or floating on the mix (type III only). The texture shall be creamy and free from coarseness, lumps, and graininess. The color of style A product shall be white to light cream and the style B shall be a light to medium shade of chocolate brown. The color shall be pleasing, uniform, and natural for fluid dairy products. The type III product (before reconstitution) shall possess a fresh, sweet pleasing flavor and aroma characteristic of the applicable style. Style A shall have a white or light cream color. Style B product shall have a light to medium shade chocolate brown color. The dehydrated mix shall be free from dirt, insect parts, hair, wood, glass, metal, or other foreign material and shall not be adulterated by the addition of preservatives. The type III product shall be free from lumps that do not fall apart under light pressure and shall be readily reconstituted in water at a temperature of 40° to 60°F when agitated with a wire wisk or by other mechanical means (see 5.4.1). The type IV product shall meet the applicable requirements for types I, II, and III and in addition shall have a clean, distinct lactic acid taste associated with a commercial type custard-like yogurt. The mix shall have a pleasing nutty-like (acetaldehyde) sweet taste, but shall not mask the basic acid character of the product or vice versa. Style A product shall be white to light cream in color. Style C product shall have a color typical for the particular flavor and shall be of the flavor specified with sufficient flavor intensity so as to be readily and accurately identified. There shall be no off flavor such as but not limited to bitter, yeasty, or diacetyl. The product shall have a smooth creamy body and shall have a light to moderate viscosity similar to that of high quality ice milk mix. In addition to all of the above requirements, the products shall meet the analytical requirements described in table I.

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TABLE I. Analytical requirements

Item	Requirement	Type I Style A	Type I Style B	Type II 1/ Style A	Type II 1/ Style B	Type III Style A	Type III Style B	Type IV Style A	Type IV Style C
1	Milk or vegetable fat (as applicable), minimum (percent)	3.25	2.75 2/	3.25	2.75 2/	12.40	10.8 3/	1.50	1.50 4/
2	Nonfat milk solids, minimum (percent) 5/	12.50	12.00	13.00	12.00	38.50	34.00	12.00	12.00
3	Moisture, maximum (percent)	-	-	-	-	2.00	2.00	-	-
4	Standard plate count (per gram), maximum	50,000	50,000	50,000	50,000	50,000	50,000	-	-
5	Coliform, count (per gram), maximum	10	20	10	20	10	20	10	10
6	Salmonellae	-	-	-	-	Negative 6/	Negative 6/	-	-
7	Yeast and mold count/gram (maximum)	-	-	-	-	-	-	10	10
8	pH range	-	-	-	-	-	-	4.0 to 4.4	4.0 to 4.3
9	Scorched particles, maximum (milligrams), per 25 gram sample	-	-	-	-	15.0	-	-	-

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TABLE I. Analytical requirements - Continued

Item	Requirement	Type I Style A	Type I Style B	Type II 1/ Style A	Type II 1/ Style B	Type III Style A	Type III Style B	Type IV Style A	Type IV Style C
10	Oxygen in head-space gas, maximum (percent) 7/ 8/	-	-	-	-	3.0	3.0	-	-
11	Vitamin A palmitate, I.U./lb of dry mix	-	-	-	-	11,200 I.U.	11,200 I.U.	-	-

1/ The minimum total fat content required by the state law where the product is to be consumed will apply. Where the state does not specify a minimum fat content, the requirements of this specification will apply.

2/ Exclusive of fat from chocolate or cocoa flavoring. The minimum total fat content shall be 3.25 percent. Plant records shall be maintained and made available upon request as an aid in determining milk or vegetable fat compliance.

3/ Exclusive of fat from chocolate or cocoa flavoring. The minimum total fat content shall be 12.00 percent. Plant records shall be maintained and made available upon request as an aid in determining milk or vegetable fat compliance.

4/ The flavor dilution factor shall be considered in the formulation so that the finished product meets the test requirements.

5/ Plant records shall be reviewed to determine compliance.

6/ Negative in a 25 gram sample.

7/ Maximum value 7 days after packaging.

8/ Sample lot average shall not exceed 3.0 percent oxygen, with no individual sample exceeding 4.5 percent.

3.3.1 Frozen product requirements. Requirements for the frozen products shall be as specified in 3.3.1.1 through 3.3.1.3.

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3.3.1.1 Types I, III (after reconstitution - see 5.4.1), style A, soft serve ice milk and type IV. The liquid mix when frozen in a soft serve type freezer to a drawing temperature of 18° to 22°F shall result in a soft serve product complying with the following characteristics:

a. No difficulty in obtaining an overrun of 30 to 40 percent for types I and III, and 40 to 60 percent for type IV which may be calculated by the following formula:

$$\frac{A-B}{B} \times 100 = \text{Percent overrun}$$

where A = weight of the mix (for convenience, use a straight sided, flat rimmed cup of approximately 8 oz. capacity and fill to the brim), and
B = weight of same volume of frozen product.

b. The color shall be white to light cream for types I, III, style A and type IV, style A, appropriate color for the particular fruit flavor of type IV, style C, and have a dry, non-glistening surface appearance.

c. The product as it emerges from the freezer shall have good peaking (coiling up) characteristics.

d. The body shall be stiff and firm and have good resistance to melting.

e. The texture shall be smooth and free from palpable ice crystals and the product shall show no tendency toward churning in the freezer cylinder.

f. The flavor shall be sweet, clean, and characteristically that of vanilla (type I, style A and type III, style A).

g. Type IV, style A shall have a distinct lactic acid taste with a characteristic vanilla flavor.

h. For type IV, style C, the flavor shall reflect the flavor character described in 3.3. Style C product may or may not show pieces of fruit in the frozen form, as applicable.

3.3.1.2 Types I and III (after reconstitution - see 5.4.1), style B, soft serve ice milk. The liquid mix when frozen in a soft serve type freezer to a drawing temperature of 18° to 22°F shall result in a soft serve product which complies with the same requirements specified in 3.3.1.1, except for the following differences:

a. The color shall be light to medium chocolate brown and the flavor characteristic of a clean and distinctive chocolate flavor, without interfering flavor-carrying enhancers, such as honey, malt syrup or molasses. In addition, the chocolate shall not have an objectionable harsh or bitter background.

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b. The frozen product shall be free of "flecking," a defect which may be described as a noticeable agglomeration of cocoa fat particles.

3.3.1.3 Types II and III (after reconstitution - see 5.4.1), direct draw milk shakes. The liquid mix when frozen in a soft serve type freezer adapted for making milk shakes or in ready-to-serve shake make freezers to a drawing temperature of 27° to 30°F shall result in a milk shake complying with 3.3.1.1 (b) and (f) for style A and 3.3.1.2 (a) and (b) for style B and with the following characteristics:

a. No difficulty in obtaining an overrun of 20-30 percent, calculated in accordance with 3.3.1.1.

b. The resulting shake shall be moderately smooth, foamy, and thick (see 6.1).

c. There shall be no evidence of component separation, gumminess, stickiness, graininess, or butteriness.

d. The product shall not leave a thick slimy coating in the mouth.

e. Additionally, type III product shall possess the following physical characteristics when the reconstituted mix is blended with the addition of water in the suggested ratios:

Texture condition desired	Ratio by volume of reconstituted mix to water
Smooth and free from palpable ice crystals	no dilution
Slightly coarse	3 to 1
Medium coarse	3 to 2
Coarse with small palpable ice crystals	1 to 1

3.4 Age of product. Types I, II, and IV products shall be delivered within 96 hours after date of manufacture. If type IV is in a frozen state, product shall be delivered within 10 days after date of manufacture. Unless otherwise specified by the procuring agency (see 6.2), type III product shall be shipped within 60 days subsequent to production.

3.5 Delivery temperature of types I, II, and IV products. The temperature of the product(s) at the time of delivery shall not exceed 45°F. In the case of type IV product in frozen form, the temperature of the product at any point shall not exceed 0°F.

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3.6 Plant qualification. The product shall originate, be processed, handled, packaged, and stored in plants meeting the requirements of MIL-STD-668 or in plants which operate under the Grading and Inspection, General Specifications for Approved Plants and Standards for Grades of Dairy Products and approved for the finished product described in this document. When Grade A milk products are required for type IV product manufacture, the plant shall have received prior sanitary approval by the Military Veterinary Services in accordance with the Grade "A" Pasteurized Milk Ordinance or shall have a pasteurization plant compliance rating of 90 or more or certified by a State Milk Sanitation rating officer and listed in the document titled, Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers, published quarterly by the U.S. Public Health Service.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA, AMS: Dairy Division, shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA, AMS: Dairy Division, for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA, AMS: Dairy Division. The USDA, AMS: Dairy Division, will determine the degree of inspection necessary to assure compliance with the requirements of this specification.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.1 and 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to

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determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.3.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can exterior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.3.3 Net weight examination, type III only (see 5.1.1). The cans of product shall be examined for net weight. The net weight of the filled and sealed cans shall be determined by weighing each sample unit on a suitable scale tared with a representative empty can and lid. Report the results to the nearest 1/8 ounce. Any nonconformance shall be classified as a minor defect. The lot size shall be expressed in units of cans. The sample unit shall be one can of product. The inspection level shall be S-3 and the acceptance quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for minor defects. In addition, the lot shall be rejected if the average net weight of the samples is less than the specified net weight.

4.3.4 Product examination.

4.3.4.1 Product characteristics examination, type III only (prior to reconstitution as specified in 5.4.1). The dehydrated product shall be examined for the defects listed in table II. The lot size shall be expressed in cans. The sample unit shall be the contents of one filled and sealed can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects.

TABLE II. Product defects, type III only 1/

Category	Defect
Major	
101	Not a fresh, sweet, pleasing flavor and aroma
102	Presence of lumps that do not fall apart under light pressure
103	Not a white or light cream color (style A)
104	Not a light to medium shade chocolate brown color (style B)

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, or metal), foreign odor or flavor (for example, stale, sour, rancid, or moldy), or foreign color shall be cause for rejection of the lot.

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4.3.4.2 Product characteristics examination, types I, II, III (after reconstitution - see 5.4.1) and IV fluid mixes. The fluid soft serve, milk shake, and yogurt mix shall be examined for the defects listed in table III. The lot size shall be expressed in primary containers. The sample unit shall be the contents of one filled and sealed primary container (except examination of type III reconstituted product shall be performed using the cans selected for table II). The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects.

TABLE III. Product defects, types I, II, III (reconstituted - see 5.4.1) and IV fluid mixes 1/ 2/ 3/

Category	Defect
Major	Applicable to types I, II, III (reconstituted) and IV
101	Not type and style specified
102	Flavor not clean, pleasant, creamy, moderately sweet, and not characteristic of vanilla and chocolate flavor types used in commercial operations (a moderately cooked flavor shall not be scored as a defect)
103	Liquid mix not viscous, smooth, readily flowable in soft serve freezer hoppers or equivalent to fresh heavy sweet cream
104	Texture not creamy
105	Presence of coarseness, lumps, or graininess
106	Not a white or light cream color (style A)
107	Not a light to medium shade of chocolate brown (style B)
108	Evidence of "fish-eyes" or undissolved stabilizer or emulsifier, applicable to type III only
	Applicable to type IV 4/
109	Does not have a clean, distinct lactic acid taste typical of custard like yogurt
110	The pleasing, nutty-like (acetaldehyde) sweet taste masks the basic lactic acid character or vice versa
111	Color not typical of the flavor specified (style C)
112	Flavor not typical or of sufficient intensity to be readily and accurately identified (style C)

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1/ The presence of foreign material (for example, insect parts, hair, wood, glass, or metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, or moldy), or foreign color shall be cause for rejection of the lot.

2/ The presence of any defect such as ropy, curdled, whey-off, watery, foamy, doughy, etc. shall be cause for rejection of the lot.

3/ Type III product not readily reconstituted shall be cause for rejection of the lot.

4/ Off-flavors such as, but not limited to, bitter, yeasty, or diacetyl, shall be cause for rejection of the lot.

4.3.4.3 Product characteristics examination of types I and III (style A and B) as frozen soft serve ice milk. The soft serve ice milk shall be examined for the defects listed in table IV. The lot size shall be expressed in primary containers. Examination shall be performed using the samples and acceptance criteria for table III.

TABLE IV. Product defects, types I and III (reconstituted - see 5.4.1), after freezing of both styles A and B soft serve ice milk mix

Category	Defect
Major	Applicable to styles A and B
101	Overrun not as specified (see 3.3.1.1)
102	Does not possess a good peaking (coiling up) characteristic
103	Not a stiff and firm body
104	Not resistant to melting
105	Not a smooth texture
106	Presence of palpable ice crystals
107	Shows tendency toward churning in freezer cylinder
108	Does not possess a dry, non-glistening surface appearance
	Applicable to style A only
109	Color not white to light cream color
110	Not a clean, sweet flavor characteristic to that of vanilla

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TABLE IV. Product defects, types I and III (reconstituted - see 5.4.1), after freezing of both styles A and B soft serve ice milk mix - Continued

Category	Defect
Major	Applicable to style B only
111	Color not a light to medium chocolate brown
112	Not a clean and distinctive chocolate flavor
113	Flavor is harsh, objectionable, or bitter
114	Presence of flecking (see 3.3.1.2 (b))

4.3.4.4 Product characteristics examination of types II and III (after freezing for direct draw milk shakes). The direct draw milk shake product shall be examined for the defects listed in table V. The lot size shall be expressed in primary containers. Examination of type II shall be performed using the samples selected for table III. Examination of type III shall be performed by selection of additional samples. The sample unit shall be the contents of one filled and sealed primary container. The inspection level shall be S-2 and the AQL, expressed as defects per hundred units, shall be 1.5 for major defects.

TABLE V. Product defects, types II and III (reconstituted - see 5.4.1), after freezing for direct draw milk shakes

Category	Defect
Major	
101	Overrun not as specified (see 3.3.1.3)
102	Shake not moderately smooth, foamy, or thick
103	Color not white to light cream color (see 3.3.1.1b) (style A only)
104	Color not light to medium chocolate brown (see 3.3.1.2a) (style B)
105	Not a clean and distinctive vanilla flavor (see 3.3.1.1f) (style A)
106	Flavor not characteristic of a clean and distinctive chocolate flavor (see 3.3.1.2a) (style B)
107	Presence of flecking (see 3.3.1.2b) (style B)
108	Evidence of separation, gumminess, stickiness, graininess, or butteriness
109	Product leaves a thick slimy coating in the mouth

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TABLE V. Product defects, types II and III (reconstituted - see 5.4.1), after freezing for direct draw milk shakes - Continued

Category	Defect
Major	Applicable to type III
110	Texture not as specified in accordance with the ratio dilution factors of 3.3.1.3f

4.3.4.5 Product characteristics examination of frozen soft serve yogurt (type IV). The frozen soft serve yogurt shall be examined for the defects listed in table VI. The lot size shall be expressed in primary containers. Examination shall be performed using the samples and acceptance criteria for table III.

TABLE VI. Product defects for frozen soft serve yogurt (type IV) 1/

Category	Defect
Major	
101	Not type and style specified
102	Flavor not clean, pleasant, creamy, moderately sweet, and not characteristic of vanilla and chocolate flavor types used in commercial operations (a moderately cooked flavor shall not be scored as a defect)
103	Texture not creamy
104	Does not have a clean, distinct lactic acid taste typical of custard-like yogurt
105	The pleasing, nutty-like (acetaldehyde) sweet taste masks the basic lactic acid character or vice versa
106	Not a white or light cream color (style A)
107	Color not typical of the flavor specified (style C)
108	Flavor not typical or of sufficient intensity to be readily and accurately identified

1/ Off flavors such as, but not limited to, bitter, yeasty, or diacetyl, shall be cause for rejection of the lot.

4.3.5 End item testing. The finished product shall be tested, as applicable, for oxygen in the headspace, fat, moisture, pH (type IV), scorched particles (type III, style A only), standard plate count, coliform count, yeast and mold (type IV), and for the presence of salmonellae in accordance with the

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requirements of 3.3. Acceptance criteria for nonfat milk solids shall be determined by review of plant records. Test procedures shall be as specified in 4.4. Test requirements for fat, oxygen, and moisture and for microbiological content shall be applicable to the sample unit. The lot size shall be expressed in terms of primary containers. The sample unit shall be one filled and sealed primary container. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units for each characteristic except for salmonellae determination, shall be 2.5. A nonconformance to a test requirement shall be considered a major defect. The sample unit for testing for scorched particles shall be approximately a 1-pound composite derived from the sample units used for unit testing. Results shall be reported to the nearest milligram for scorched particles, to the nearest 0.1 percent for fat and headspace oxygen, and as positive or negative for salmonellae. All microbiological counts shall be reported in accordance with Standard Methods for the Examination of Dairy Products. The finding of any test as positive for salmonellae shall be cause for rejection of the involved lot of finished product. In addition, nonconformance to footnotes 1/, 2/, 3/, 4/ and 5/ of table I shall be classified a major defect and shall be cause for rejection of the lot.

4.3.6 Packaging inspection.

4.3.6.1 External examination of cans and can leakage. External examination of cans and can leakage shall be performed in accordance with the applicable examination criteria of PPP-C-29.

4.3.6.2 Examination of exterior can coating (when applicable). Examination of exterior can coating shall be performed in accordance with the applicable examination criteria of TT-C-495.

4.3.6.3 Examination of can labeling. Examination of can labeling shall be in accordance with the examination criteria of MIL-L-1497.

4.3.6.4 Shipping container examination. Examination of the filled, closed, and reinforced shipping containers shall be made in accordance with the applicable examination criteria in the appendix to PPP-B-636. In addition, the following defects shall be included in the examination: Major: Nomenclature, NSN, contract number, and date of pack missing, incorrect, or illegible; Minor: (1) Other marking missing, incorrect, or illegible. (2) Arrangement or number of cans not as specified. (3) Sleeve missing. For level B pack, when applicable, shipping containers reinforced with other than adhesive, filament reinforced tape shall not be acceptable. When packing for level C is required to be in accordance with 5.2.3, examination shall be in accordance with PPP-B-636 except that only the defects for marking and arrangement or number of cans shall apply.

4.3.6.5 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

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4.3.6.6 Examination of age of product at time of shipment. Examination shall be made at time of shipment to determine compliance with 3.4. This determination shall be ascertained by examination of records, codes, marked date of pack, or other valid means. Nonconformance to the above referenced requirements or age marking missing, incorrect, or illegible shall be cause for rejection of the lot.

4.3.6.7 Examination of temperature at time of delivery (types I, II, and IV). Examination shall be made at time of delivery to determine compliance with the temperature requirements of 3.5. Nonconformance to the temperature requirements as indicated by an accurate temperature measuring device is a critical defect and shall be cause for rejection of the lot.

4.4 Methods of inspection.

4.4.1 Chemical analyses. Chemical analyses shall be made in accordance with the methods and publications listed below:

Test	Source	Method
Ingredient:		
Coconut fat and vegetable fat, as applicable:		
Moisture and volatile matter	1/	Ca 2b-38
Melting point (Wiley method)	1/	Cc 2-38
Free fatty acid	1/	Ca 5a-40
Peroxide value	1/	Cd 8-53
Iodine value	1/	Cd 1-25
Color	1/	Cc 13b-45
Fat stability (AOM)	1/	Cd 12-57
Finished product:		
Fat (type I, II and IV product)	2/	Roese-Gottlieb (Official)
Fat (type III product)	2/	Fat in Dried Milk
Moisture	3/	Dry Whey
Scorched particles	4/	Scorched Particles (25 gram sample)
Oxygen in headspace	5/	Determination of Oxygen and Carbon Dioxide Content Gases in Hermetically Sealed Gas Packed Containers
pH	6/	Hydrogen Ion concentration

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1/ Official Methods and Recommended Practices of the American Oil Chemists' Society, Section C, Commercial Fats and Oils.

2/ Official Methods of Analysis of the Association of Official Analytical Chemists for types I, II and IV; - chapter: Dairy Products; section: Ice Cream and Frozen Desserts for type III; chapter: Dairy Products; section: Dried Milk, Nonfat Dry Milk and Malted Milks.

3/ Methods of Laboratory Analysis, Dairy Division, USDA, DA Instruction Series 918-109-2.

4/ American Dairy Products Institute, Standards for Grades of Dry Milks, Including Methods of Analysis, Bulletin 916.

5/ Methods of Laboratory Analysis, Dairy Division, USDA, DA Instruction Series 918-103-10.

6/ Standard Methods for the Examination of Dairy Products, Chapter: Chemical and Physical Methods, Section: Acidity.

4.4.2 Microbiological analyses. Microbiological analyses shall be performed in accordance with the applicable methods for specific dairy product components described in Standard Methods for the Examination of Dairy Products and for the finished product noted as follows:

Test	Method
Standard plate count (types I and II)	Condensed milk 1/
Standard plate count (type III)	Dry milk 1/
Coliform count (types I and II)	Test for coliform group in condensed milk 1/
Coliform count (type III)	Test for coliform group in dry milk 1/
Yeast and mold count (type IV)	Test for yeast and mold count 2/
Coliform count (type IV)	Test for coliforms 2/
Salmonellae test	3/

1/ Chapter: Microbiological Methods for Concentrated Milk and Dry Milk.

2/ Chapter: Microbiological Methods for Cheese and Cultured Products.

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3/ U.S. Department of Health and Human Services, Bacteriological Analytical Manual, Dried Milk Section.

4.4.3 Tin coating weight. Tin coating weight shall be determined by any method specified in PPP-C-96.

5. PACKAGING

5.1 Preservation. Type III product shall be preserved in accordance with level A or level C, as specified (see 6.2). Type I, II, and IV product shall be preserved in accordance with level C.

5.1.1 Level A. A net weight of 5 pounds of type III product shall be filled into a size 603 by 812, or 603 by 700 as required, open-top style, round, metal can with soldered or welded side seam and compound-lined, double-seamed ends. The No. 10 can (603 by 700) or the No. 12 can (603 by 812) shall be optional for type III, class 1 product, but the No. 12 can must be used for type III, class 2 product to accommodate the lower bulk density in the product with vegetable fat. A minus 3/8-ounce tolerance will be allowed in any one container provided the average net weight of the cans inspected in accordance with 4.3.3 is not less than 5 pounds. The can shall be made throughout from commercial 0.25-pound per base box electrolytic tin plate (see 4.4.3), and shall be coated overall outside with coating conforming to type I or, when specified (see 6.2), type III of TT-C-495. The product shall be gaspacked by removing the air from the filled can and replacing it with nitrogen. The filled and sealed can shall not leak when tested in accordance with 4.3.6.1.

5.1.2 Level C.

5.1.2.1 Type I, II, or IV. Type I, II, or IV product shall be preserved in a manner to protect it from damage or deterioration during shipment from the supply source to the first consuming activity.

5.1.2.2 Type III. Type III product shall be preserved in accordance with 5.1.1, except that cans with or without commercial exterior coating will be acceptable. Alternatively, cans may be made from 0.20 pound per base box electrolytic tinplate that is coated with a commercial exterior coating.

5.2 Packing. Type III product shall be packed in accordance with level A, B, or C, as specified (see 6.2). Type I, II, or IV product shall be packed in accordance with level C.

5.2.1 Level A packing. Six cans of type III product, preserved as specified in 5.1, shall be packed on end, three in length by two in width in a snug-fitting fiberboard shipping container constructed, closed, and reinforced in accordance with style RSC, grade V2s of PPP-B-636. Alternatively, the shipping container shall be a wax-impregnated fiberboard box constructed in accordance with style RSC, class 1, type SWCFI, grade 200 of PPP-B-1163. Closure of the wax-impregnated box shall be with hot melt adhesive in accordance with the appendix of PPP-B-636. All shipping containers shall be reinforced with

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nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. The reinforcing bands shall be uncolored or of a color which contrasts with black. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Six cans of type III product, preserved as specified in 5.1, shall be packed on end, three in length by two in width in a snug-fitting fiberboard shipping container constructed, closed, and reinforced in accordance with style RSC, grade V3c, V3s, or V4s of PPP-B-636. Alternatively, the shipping container shall be a wax-impregnated fiberboard box constructed in accordance with style RSC, class 1, type SWCFI, grade 200 of PPP-B-1163. Closure of the wax-impregnated box shall be with hot-melt adhesive in accordance with the appendix of PPP-B-636. All shipping containers shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. The reinforcing bands shall be uncolored or of a color which will contrast with black.

5.2.3 Level C packing. Six cans of type III, preserved as specified in 5.1, shall be packed on end, three in length by two in width in a snug-fitting fiberboard shipping container. The shipping container for all types shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable, except that the closure of fiberboard boxes shall be in accordance with the appendix of PPP-B-636.

5.3 Unit loads. When specified (see 6.2), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling and marking.

5.4.1 Cans. Cans of the product shall be labeled in accordance with MIL-L-1497. Labeling shall include directions for use as follows:

FOR TYPE III, STYLE A

Reconstitution for producing a soft serve fluid mix shall be in the proportion of 5 pounds of dehydrated product to 5 quarts of 40° to 60°F potable water.

FOR TYPE III, STYLE B

Reconstitution for producing a soft serve fluid mix shall be in the proportion of 5 pounds of dehydrated product to 5 quarts of 40° to 60°F potable water.

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FREEZING INSTRUCTIONS FOR SOFT SERVE ICE MILKS (TYPES I AND III)

Freeze in soft serve type freezer to a drawing temperature of 18° to 22°F or when the product being frozen reaches a good, stiff peaking condition upon filling into container.

FREEZING INSTRUCTIONS FOR DIRECT DRAW MILK SHAKES

TYPES II AND III

Reconstitute dehydrated product as for soft service ice milk and freeze in same type freezer to a drawing temperature of 27° to 30°F or to slushy, soft consistency, depending on preference. Reconstituted mix may be diluted as follows to attain desired shake characteristics:

Texture Condition Desired	Ration by volume of reconstituted mix to water
Smooth and free from palpable ice crystals	No dilution
Slightly coarse	3 parts mix to 1 part water
Medium coarse	3 parts mix to 2 parts water
Coarse and icy with palpable ice crystals	1 part mix to 1 part water

5.4.1.1 Commercial labeling. When specified (see 6.2), commercial labeling is acceptable provided the finished product complies with all other specification requirements.

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129. In addition, the shipping container shall be marked on top in letters 7/16 inch to 1 inch (in height) with the following precautionary marking:

DO NOT STORE ABOVE 70°F (for type III product)

KEEP REFRIGERATED 32°F to 40°F (for type I, II and IV product)

KEEP FROZEN -10°F (for type IV product, if frozen)

For type IV product that is frozen include the following marking:

To thaw, place containers, with space intervals between them at refrigeration temperature (35° to 40°F) for 48 to 72 hours. Use only when completely thawed. When not in use, refrigerate at 32° to 40°F.

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Technical information. Type I, II, and IV products are classified as perishable, and type III as semiperishable; however; storage of type III in a cool location will aid in retention of flavor quality. Fresh liquid mix meeting the requirements of this specification and using good commercial practice has a shelf life of 5 days. Type IV mix when frozen has an estimated shelf life of 6 months when stored at a uniform temperature of -15° to -20°F and when the package is kept sealed. Type III product, when stored at 50° to 70°F, has an approximate shelf life of 1 year. Class 2, imitation labeled products are not intended for use in Continental United States (CONUS). It should be noted that no specific formulation for the products described in this specification is made mandatory because of a wide range in personal preference and the difficulty to define an ideal product. Several factors affect product characteristics, but basically high total solids levels and low drawing temperatures have a pronounced effect upon stiffness and dryness. A high sugar content will require a lower drawing temperature to provide a body and dryness comparable to a mix containing less sweetening but frozen to a higher drawing temperature. The degree of thickness and smoothness in a direct draw milk shake varies considerable with formulation, drawing temperature, overrun and the degree and intensity of agitation. Sufficient flexibility in the formulation should be provided, so that, when preferred, the resultant shake can be frozen (by controlling drawing temperature and overrun) to a thicker, heavier consistency or to a light, thin product. A lower total solids will contribute to a cold, icy shake sensation while a mix containing a high solids content will yield a thick, heavy bodied shake.

6.1.1 Type IV product. The distinguishing difference between yogurt and other cultured dairy products sold commercially in CONUS is the use of two different types of organisms in a 1:1 blend. Both organisms, Streptococcus thermophilus and Lactobacillus bulgaricus, supplement and stimulate one another (i.e., act in symbiosis) in terms of growth, acid, and flavor enhancement. Culture handling requires skill and aseptic technique, along with a good culture rotation program to ensure the proper balance between the organisms and to avoid airborne contamination. The flavor profile for a good yogurt product is difficult to describe; the most practical approach is to strike a balance between a desirable acid level (pH) and a pleasing and nutty-like flavor and aroma background (acetaldehyde). The difficulty becomes increasingly more apparent as the consumer's palate is masked by both the sweeteners used and the added flavors. What must be avoided is a highly sugared product and just sufficient fruit flavor should be added to permit the natural yogurt flavor to come through. The development of excess acidity (i.e., below pH 4.0) will tend to override the more exquisite yogurt flavor, only to be "covered up" by the use of extra sweeteners. The manufacture of soft serve yogurt is not presently an industry standardized procedure. The variables involved in the process are

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numerous and no attempt is made in this specification to specify a definite procedure. However, the following basic guidelines are presented in an attempt toward the recognition of an acceptable manufacturing process:

1. Dry blend all the dry ingredients (the amount of sweeteners used should not exceed 8 percent of the total called for in the formulation).

2. Disperse into the fluid dairy products to avoid lumping.

3. Pasteurize and homogenize blend in accordance with 3.2.1.

4. Cool to 105° to 110°F.

5. Inoculate with 2 percent to 3 percent active yogurt culture starter.

6. Ripen to pH of 4.5 to 4.6.

7. Cool to 80°F or lower and add remainder of sweetener. (Note: Adding sugars and flavors above 80°F will tend to induce whey separation.)

8. As cooling progresses to the 40°F level, acid development continues at a slower rate; the rate of cooling depends on temperature, size and spacing of containers, air circulation (velocity of air movement) etc.

9. The pH should now be between 4.0 and 4.4.

10. Flavor may now be added and the entire mix be subject to texturizing, using an homogenizer (no pressure) or recirculating briefly through a positive pump. An in-line texturator and screen may accomplish the same purpose.

11. The product is now ready for packaging.

12. Store at temperature below 40°F.

6.1.2 Suggested formulations. The following examples of suggested formulation have been found to produce finished products conforming to the requirements of this specification:

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Ingredients	Type I Style A (percent)	Type I Style B (percent)	Type II Style A (percent)	Type II Style B (percent)	Type III Style A (percent)	Type III Style B (percent)	Type IV Style A (percent)	Type IV Style C (percent)
Fat	3.25	3.25	3.25	3.25	12.40	12.00	1.50	1.50
Nonfat milk solids	12.50	12.00	13.00 3/	12.00 3/	38.50	34.00	13.00	12.00
Sucrose	11.70	14.60	9.00	12.00	36.00	39.00	10.0	10.00
Corn syrup solids	3.10	3.20	-	-	9.60	4/	4.0	4.00
Chocolate flavoring	-	1/	-	1/	1/	5/	-	-
Vanilla flavoring	1/	1/	1/	1/	1/	1/	1/	-
Stabilizer	2/	2/	2/	2/	2/	2/	2/	2/
Emulsifier	2/	2/	-	-	2/	2/	2/	2/
Total solids (approx.)	32.00	37.00	24.00 3/	28.00 3/	98.0	98.0	30.0	30.00

1/ Vanilla flavoring, depending on type and strength, should be used in amounts commensurate with flavor manufacturers' recommendations or in sufficient amounts to provide a pleasing consumer-acceptable flavor intensity. The use of vanilla flavoring in chocolate products is optional. Harsh flavor tones, particularly characteristic of vanillin, are to be avoided. Chocolate flavor, depending on types used, in a range of from 1.5 to 3.0 percent is generally acceptable as providing a characterizing and pleasing chocolate flavor intensity.

2/ The amount of stabilizer-emulsifier used in the liquid mix or prior to dehydration may be in the range from 0.3 to 0.5 percent.

3/ For a smooth shake drink, a nonfat milk solids content of 13.00 percent (style A) or 12.00 percent (style B) is recommended. Corresponding total solids will run about 28.00 percent (style A) or 29.00-30.00 percent (style B), respectively. For a coarse shake, solids content may be cut to less than 11.00 percent nonfat milk solids, but not generally below 8.50 percent for style A. Style B nonfat milk solids may run as low as 9.00 percent.

4/ Corn syrup solids quantity variable depending on type and quantity of chocolate flavoring used.

5/ Chocolate flavor may range from 6.0 to 8.0 percent depending on type and quantity used.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, style, and flavor, when type IV style C is required, (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When vegetable fat may be used (see 3.1.3).
e. When type III product may be delivered more than 60 days after date of production (see 3.4).
f. When type IV product is to be delivered frozen (see 3.5).
g. Levels of preservation and packing required (see 5.1 and 5.2).
h. Type of exterior can coating required (see 5.1.1).
i. Type and class of unit load when required (see 5.2.1 and 5.3).
j. When commercial labeling is acceptable (see 5.4.1.1).

6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Subject term (key word) listing.

Desserts
Ice cream
Soft serve desserts

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8910-0454)

Review activities:

Army - MD
Navy - MC
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-43717D	Ice Milk, Milk Shake and Yogurt Mixes

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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